

Table 4. Top wheat cultivars planted in Kansas by district and percent of seeded acreage.

DISTRICT 10 (NORTHWEST)		DISTRICT 40 (NORTH CENTRAL)		DISTRICT 70 (NORTHEAST)	
Jagger	23.0	Jagger	23.6	2137	32.3
TAM 107	13.5	2137	16.0	Karl/Karl 92	22.7
2137	12.9	Karl/Karl 92	12.2	Jagger	11.7
Trego-HWWW	7.5	Dominator	6.5	Dominator	7.2
AgriPro Thunderbolt	4.3	2163	1.5	2163	6.5
DISTRICT 20 (WEST CENTRAL)		DISTRICT 50 (CENTRAL)		DISTRICT 80 (EAST CENTRAL)	
TAM 110	21.1	Jagger	40.6	Jagger	36.5
Jagger	12.2	2137	18.9	2137	33.1
2137	11.5	Dominator	5.7	Karl/Karl 92	8.3
Trego-HWWW	9.6	Karl/Karl 92	3.9	2163	3.7
TAM 107	9.1	Ike	1.3	2174	2.8
DISTRICT 30 (SOUTHWEST)		DISTRICT 60 (SOUTH CENTRAL)		DISTRICT 90 (SOUTHEAST)	
Jagger	27.0	Jagger	69.8	Jagger	49.7
TAM 110	20.4	2137	8.6	2137	24.9
2137	11.6	2174	6.6	2174	7.8
Ike	9.0	AgriPro Coronado	1.4	AgriPro Coronado	2.6
TAM 107	6.3	Karl/Karl 92	1.3	Karl/Karl 92	2.1

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Update on heavy metals in soil at the Manhattan, KS, Biosolids Farm growing winter wheat.

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Last year, we reported concentrations of heavy metals in soil at the Manhattan, KS, Biosolids Farm, which grows winter wheat on the sludge-injected soil. However, the samples that we reported as ‘controls’ were not labelled correctly. These samples came from a new part of the farm that had received sludge during the summer of 2000. This area is labelled ‘Area XIV’ on the map of the sludge farm. The sludge came from a lagoon that had held the aerobically digested sludge (2 % solids or less) since 1978. The lagoon was 0.33 ha in area and 5.18 m in depth. All of this liquid was applied during the early summer of 2000 at a rate of 14.9 metric tons/ha to a field, which then grew soybeans that summer. The soil was sampled on 30 November, 2000, by personnel at the Manhattan, KS, Wastewater Treatment Plant. The soil that had received sludge yearly for 25 years and that grew winter wheat, in the old area of the farm, was sampled by the personnel on 13 March, 2001. This old area is labelled ‘Area IV’ on the map of the sludge farm, which was established in 1976. In the early years of the sludge farm (1976–92), 32 t/ha dry sludge were applied yearly. On 25 November, 1992, the EPA published regulations limiting land disposal of sludge (known as the 40 Code of Federal Regulations Part 503) and rates of sludge application now must be based on agronomically acceptable practices, which depend on the type of crop grown and its nitrogen requirements. Because we had no control samples, on 8 June, 2002, we sampled soil adjacent to Area IV. This area was fenced-off and holds the shed that houses the sludge injector when it is not in use. We now report the concentration of metals in the three different areas (Table 1).

In general, our conclusions from last year hold. That is, the results show that, after 25 years of application of biosolids to the farm, concentrations of heavy metals have not increased in the soil, except for Cu, Pb, and Zn. Lead is the only toxic heavy metal, and the reason for its elevated level is not known.